

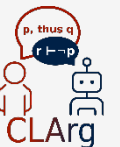
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Supported Abstract Argumentation for Case- Based Reasoning

Extended Abstract

IARML@IJCAI25

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Background

Case-Based Reasoning

- The goal of Case-Based Reasoning (CBR) is to solve new problems by comparing them with past cases.
- We can use CBR for classification of data points, framed as an analogical problem such as:

$$x_i : y_i :: x_j : y_j$$

where:

- x_i = input features of a known case
- y_i = class label of that case
- x_j = input features of the new case
- y_j = predicted label (unknown)

Background

Case-Based Reasoning

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- We can use CBR for classification of data points, framed as an analogical problem such as:

$$x_i : y_i :: x_j : y_j$$

- But what can we do when we have multiple, conflicting analogies?

New Patient

- Eats > 5 fruits daily
- Excess calories
- > 8 glasses of water daily
- High fat intake

-----**Predicted: ???**

Patient 4

- Eats > 5 fruits daily
- Excess calories
- High fat intake

-----**Unhealthy**

Patient 3

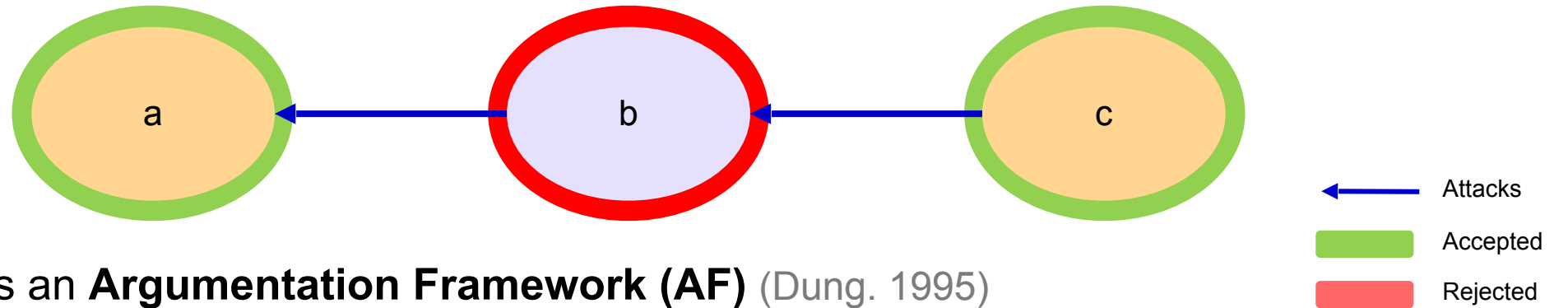
- Eats > 5 fruits daily
- Excess calories
- > 8 glasses of water daily

-----**Healthy**

Background

Abstract Argumentation

- Abstract Argumentation is a formalism for reasoning with **arguments** and their **attacks**

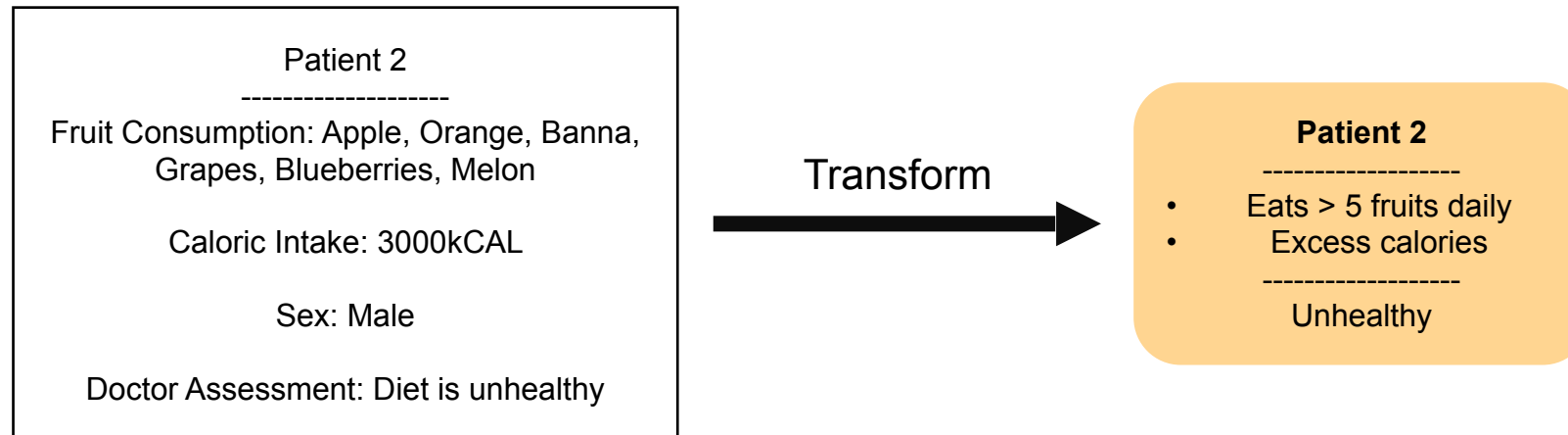


- This is known as an **Argumentation Framework (AF)** (Dung. 1995)
- We can compute which arguments are considered **acceptable** and which to **reject** by applying an argumentation **semantics** – we will focus on the **Grounded Semantics**

Background

Abstract Argumentation for Case-Based Reasoning

- AA-CBR (Cyras, et al. 2016) is an ML model that *argues* with previously observed labelled cases to predict the outcome for a new, unlabelled case
- Data points are **characterised** with some representation that we call as a **case** or **argument**

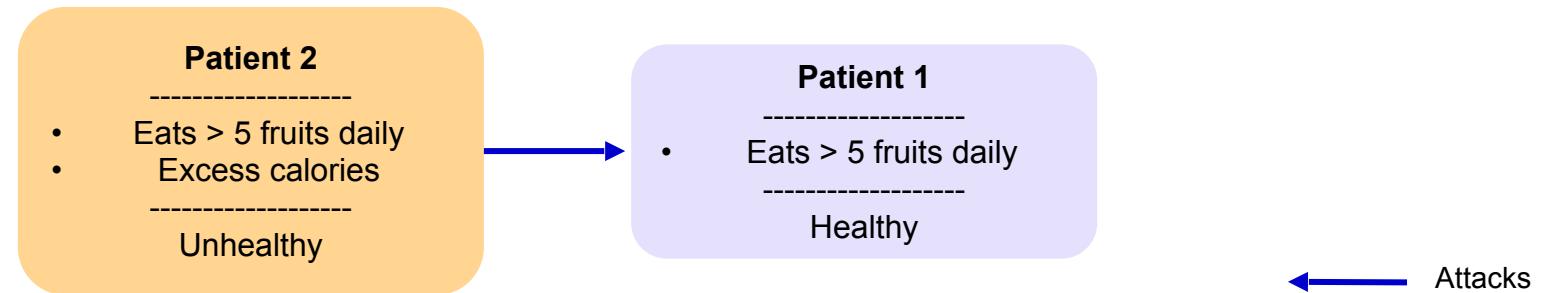


- Here we characterise each case using a **set** of binary features

Background

Abstract Argumentation for Case-Based Reasoning

- AA-CBR (Cyras, et al. 2016) is an ML model that *argues* with previously observed labelled cases to predict the outcome for a new, unlabelled case
- A case, a attacks case b iff a 's outcome differs from b 's and a is **more exceptional** than b and a is **the minimal possible attacker of b**

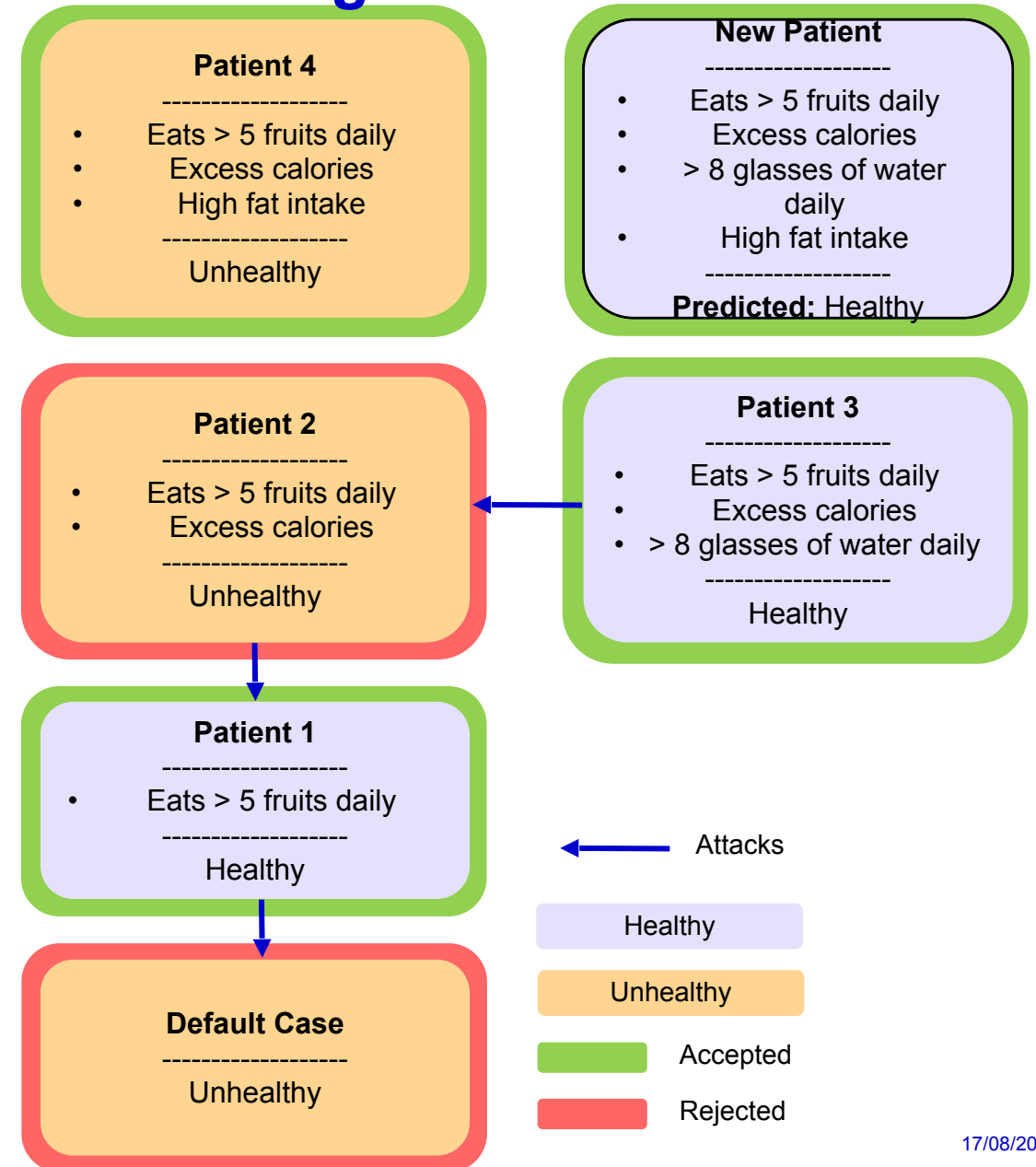


- Here, we define **exceptionality** using the **superset** relation

Abstract Argumentation for Case-Based Reasoning

Example Use Case

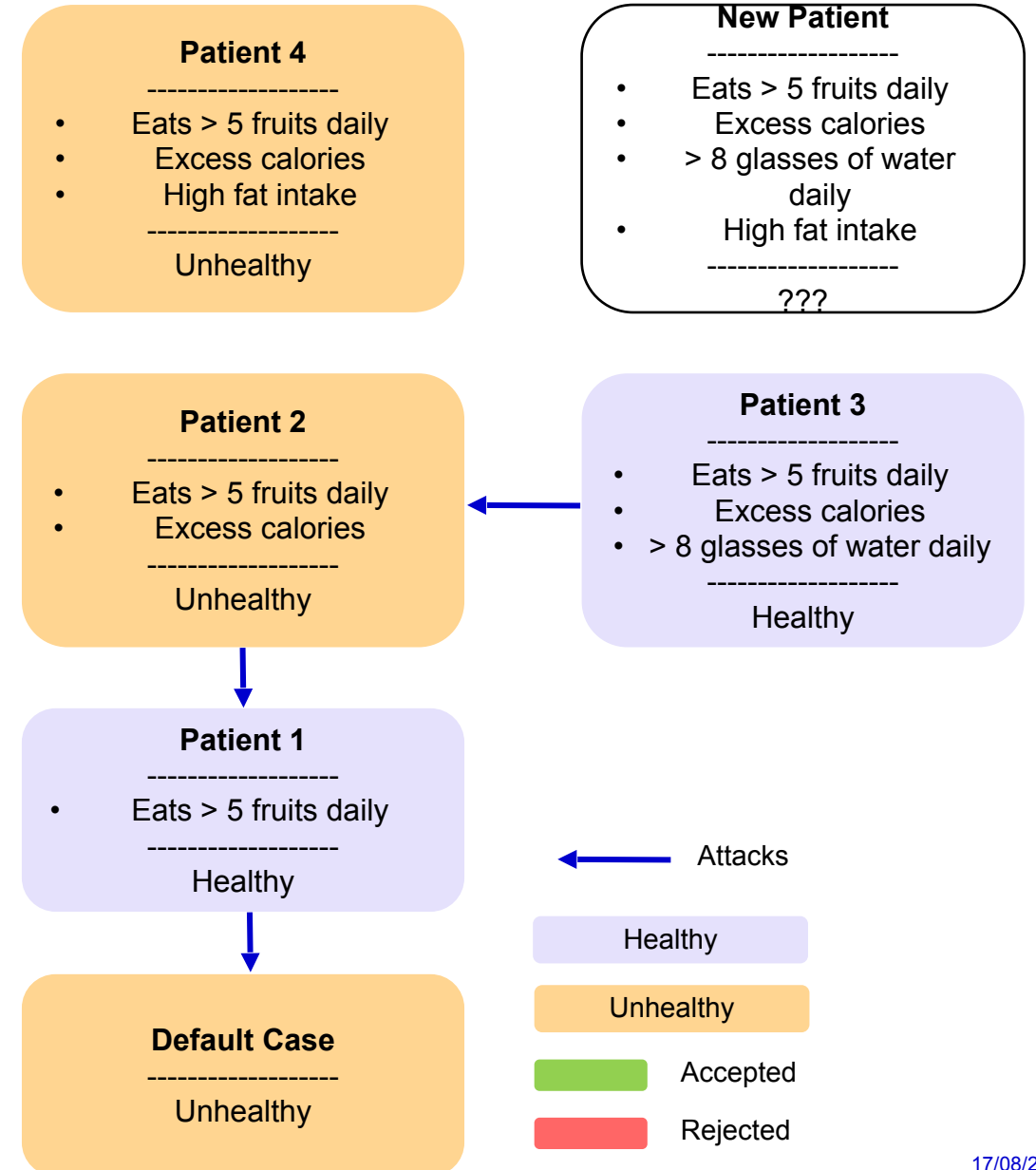
- Let's use AA-CBR to assess patient diet.
- We begin with a default case (characterised by the empty set)
 - A case with additional features and an alternative outcome attacks
 - We compute the *semantics* to determine the outcome



Abstract Argumentation for Case-Based Reasoning Problems

- The case representing Patient 4 does not contribute to the debate! We call this argument a **spike**.
- AA-CBR only accommodates **minimal** exceptions to existing rules.
- However, Patient 4 could be viewed as evidence that further **supports** the argument made by the case of Patient 2.

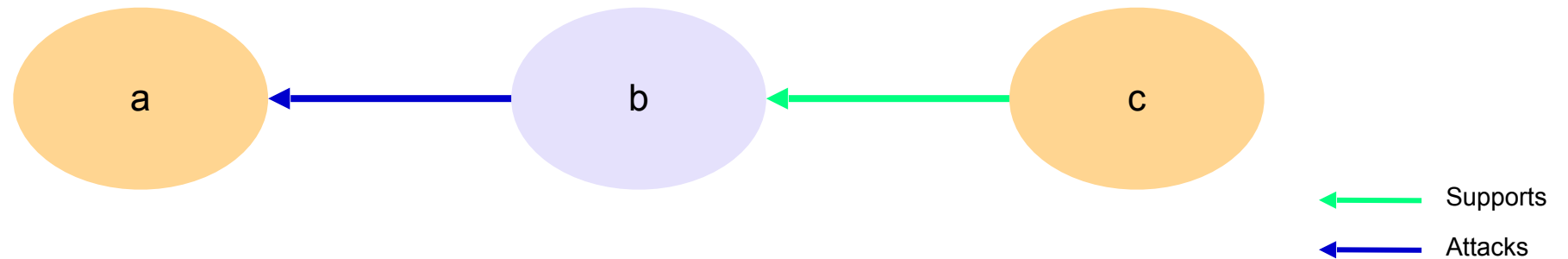
How can we accommodate this support?



Background

Bipolar Abstract Argumentation

- Bipolar Abstract Argumentation is a formalism for reasoning with **arguments** and their **attacks** and **supports** (Cayrol, et al. 2013).



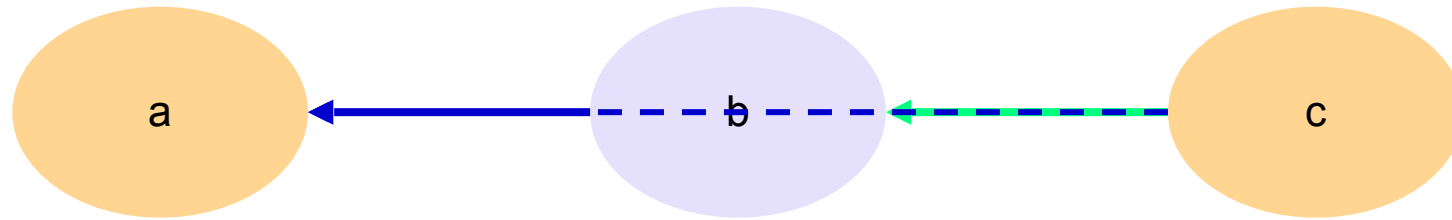
- This is known as a **Bipolar Argumentation Framework (BAF)**

Background

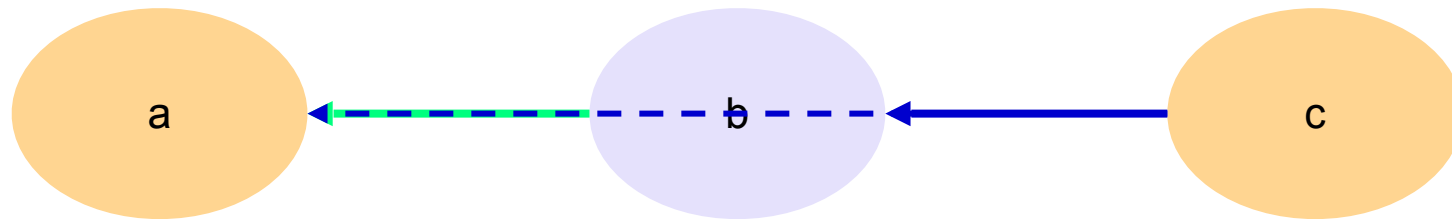
Bipolar Abstract Argumentation

- To apply argumentation semantics, we can transform a BAF into an AF (Cayrol, et al. 2013). We focus on two methods:

1. Supported Attacks:



2. Secondary Attacks:

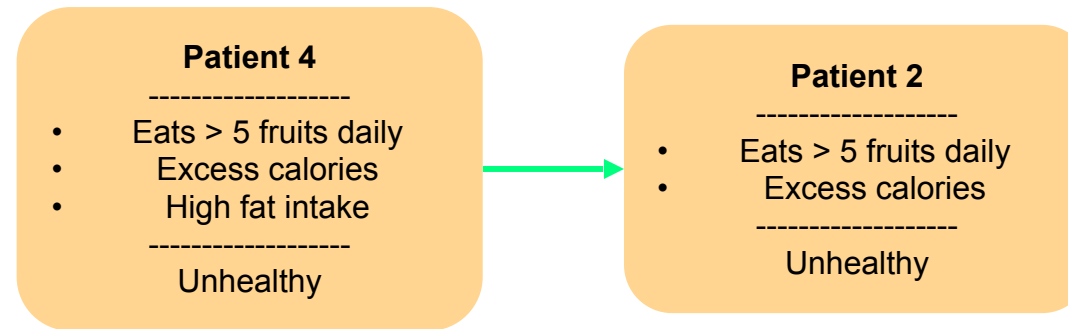


← Supports
← Attacks

Adding Supports

Supports Definition:

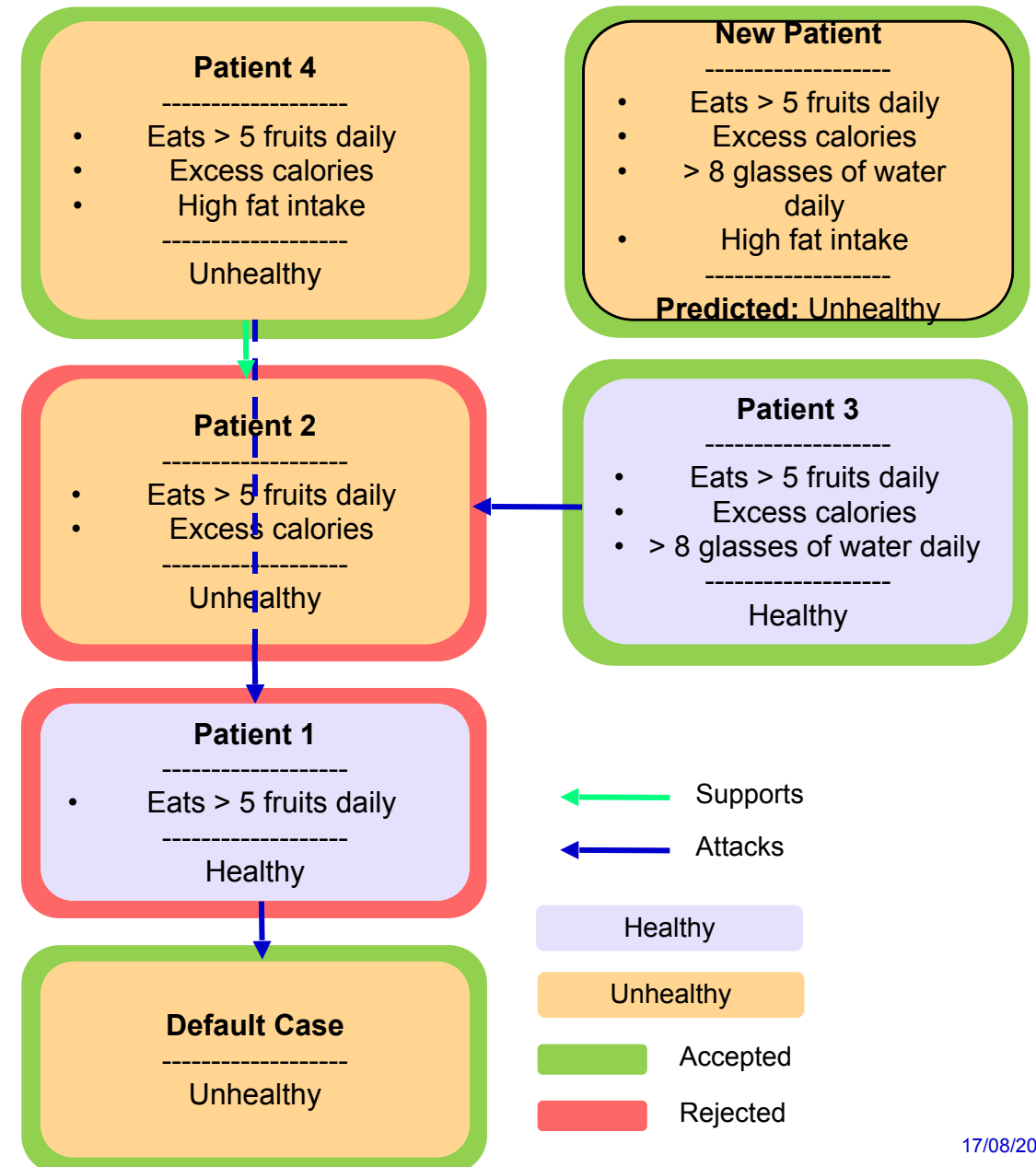
- A case, a supports case b iff a 's outcome is equal to b 's and a is **more exceptional** than b and a is **the minimal possible attacker or supporter of b**



- Note that the minimality condition gives priority to attacks – this ensures we only allow supports from spikes

Example Use Case

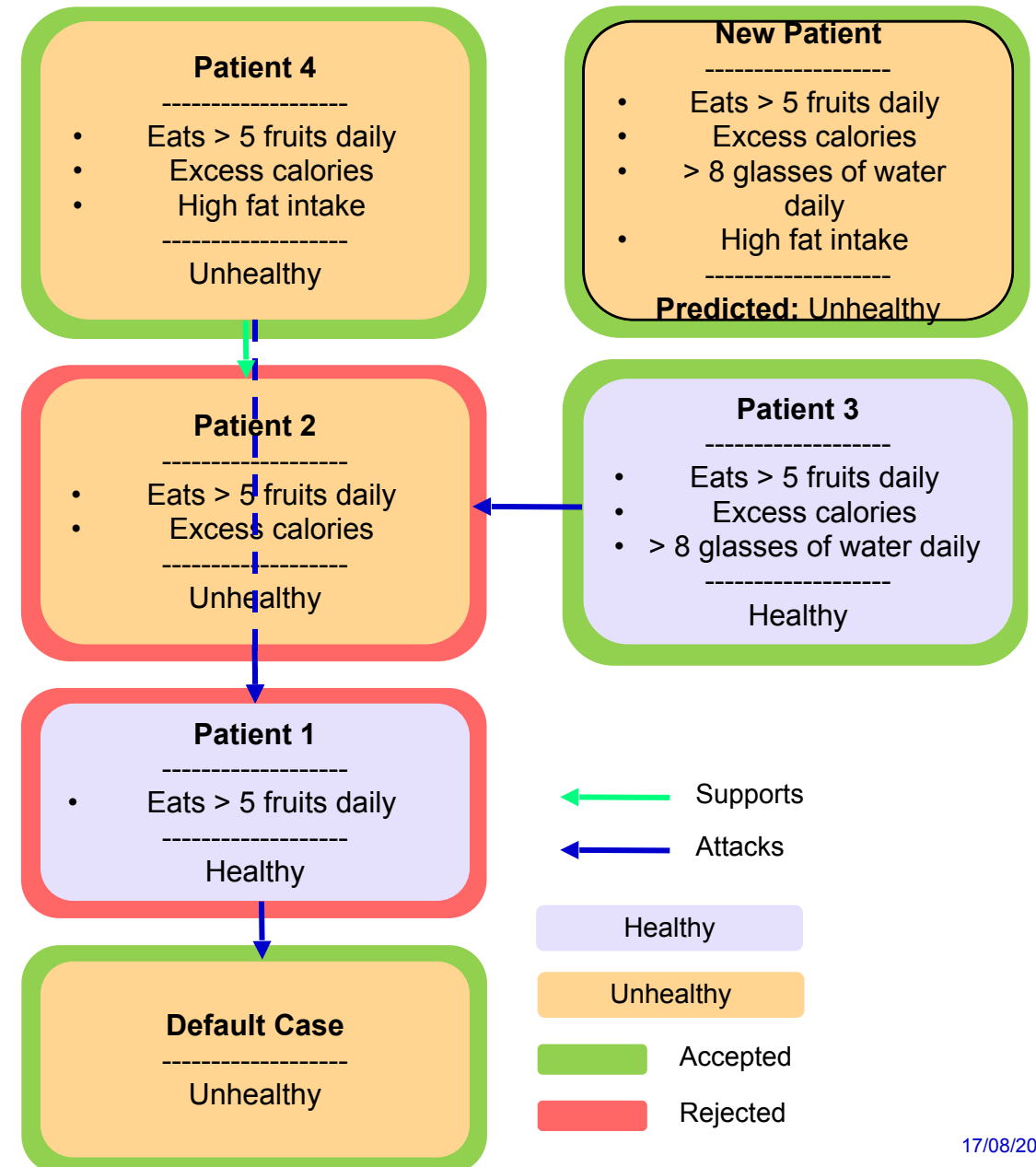
- We can add a **supports** relation from Patient 4 to Patient 2
- We can interpret this support as a supported attack
- Now when we compute the semantics, Patient 4 contributes to the debate – we can use supports to contest predictions and change incorrect ones to correct ones
- Minimality is still largely respected - we only violate the previous notion of minimality when there are spikes that otherwise would not contribute to the debate



Analogical Reasoning

Case-Based Reasoning

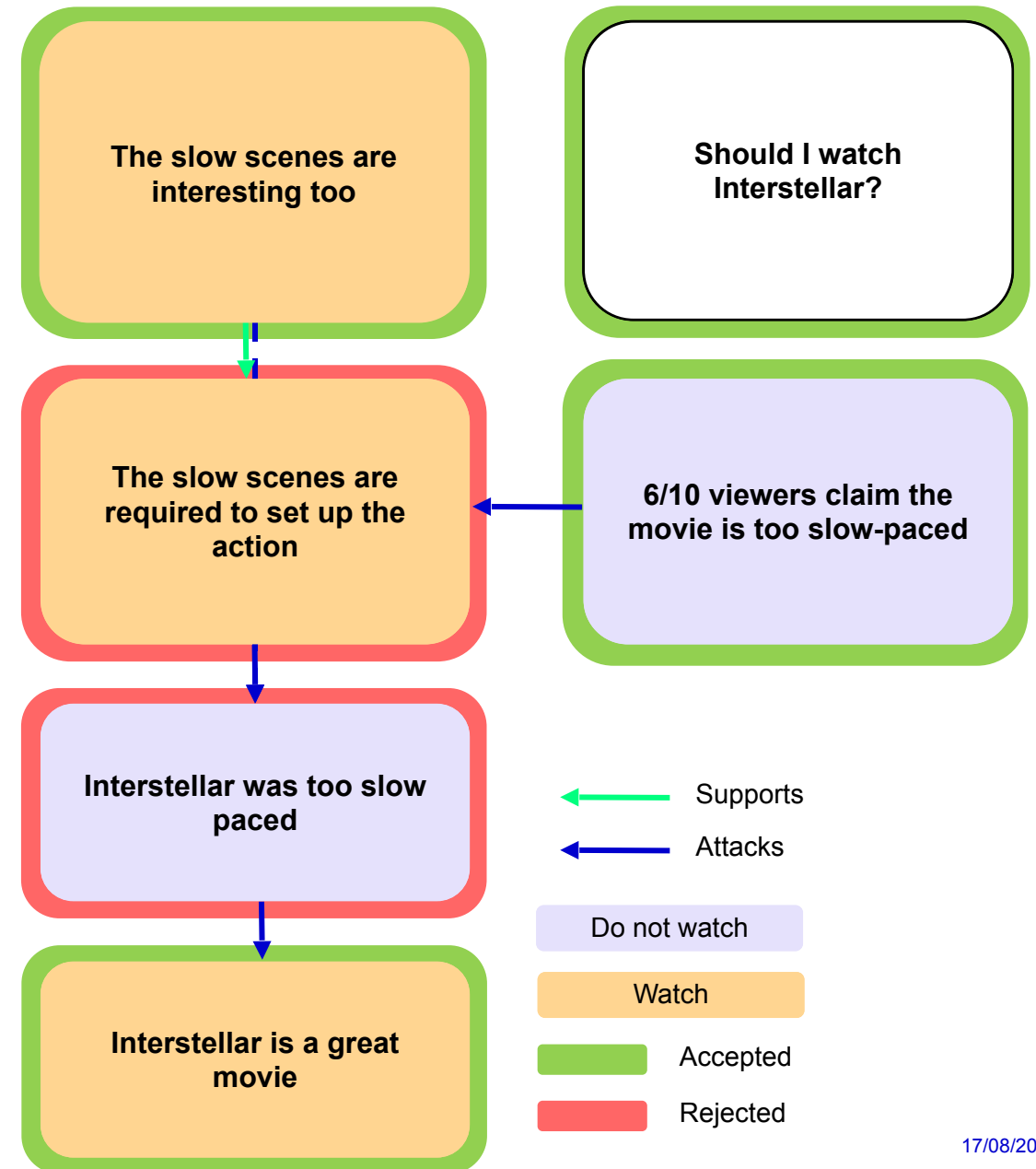
- Case-Based Reasoning (CBR) solves problems by mapping similar situations to new ones
- So each case in the casebase is a possible analogy for the predicted outcome. Argumentation helps resolve conflicts between possible analogies



Analogical Reasoning

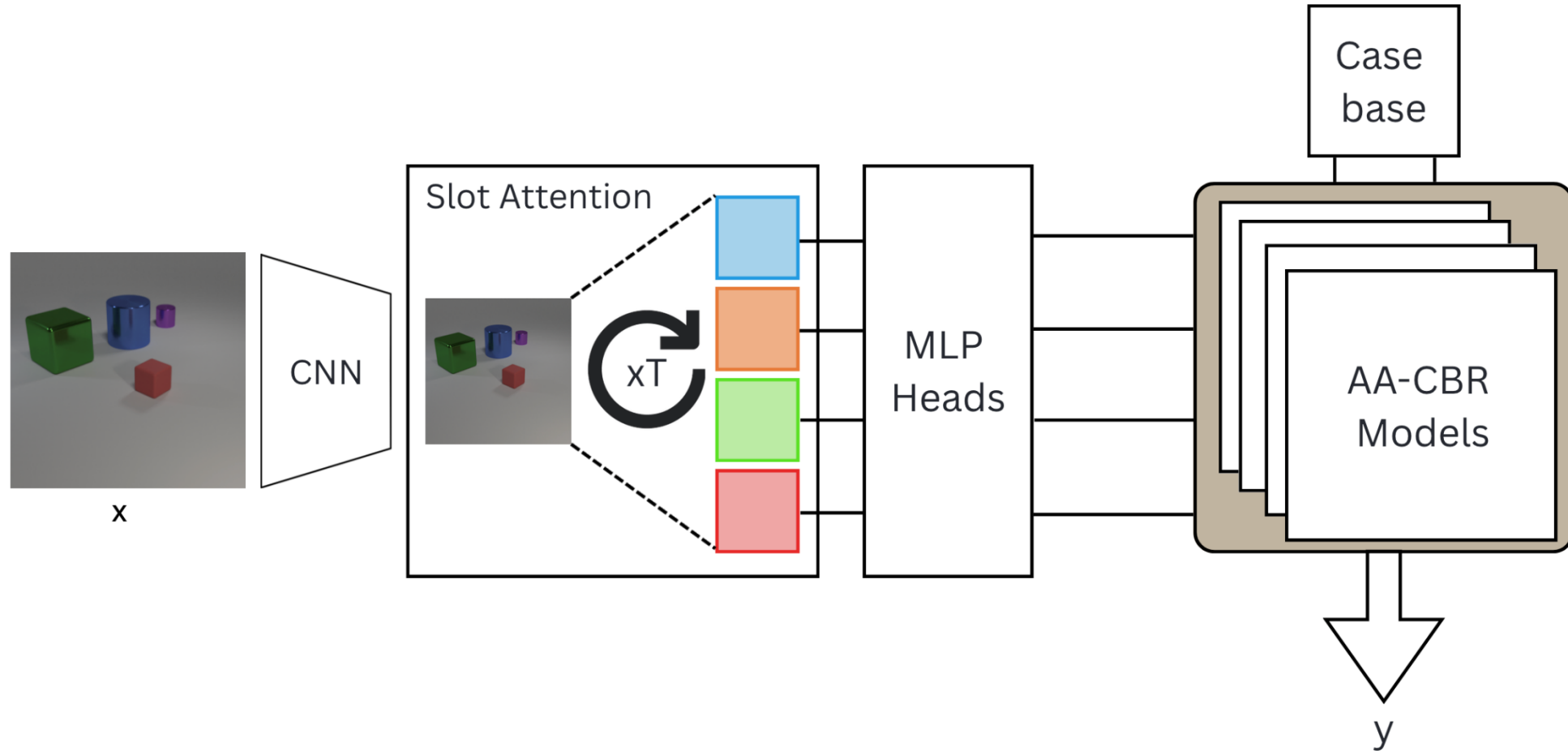
Abstract Argumentation

- Once the argumentation framework has been constructed, we apply abstract argumentation semantics
- The resolution of the debate therefore does not depend on the contents of the argument
- Abstract argumentation resembles analogical reasoning in that both are invariant under content replacement, provided relational structure is preserved.
- Unlike analogical reasoning, abstract argumentation focuses on conflict resolution rather than knowledge transfer



Supports in Practice (Preliminary work)

Slot Attention Argumentation for Case-Based Reasoning



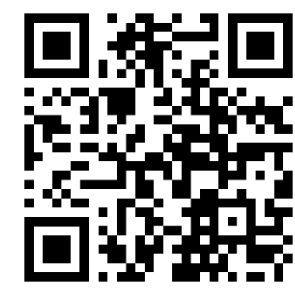
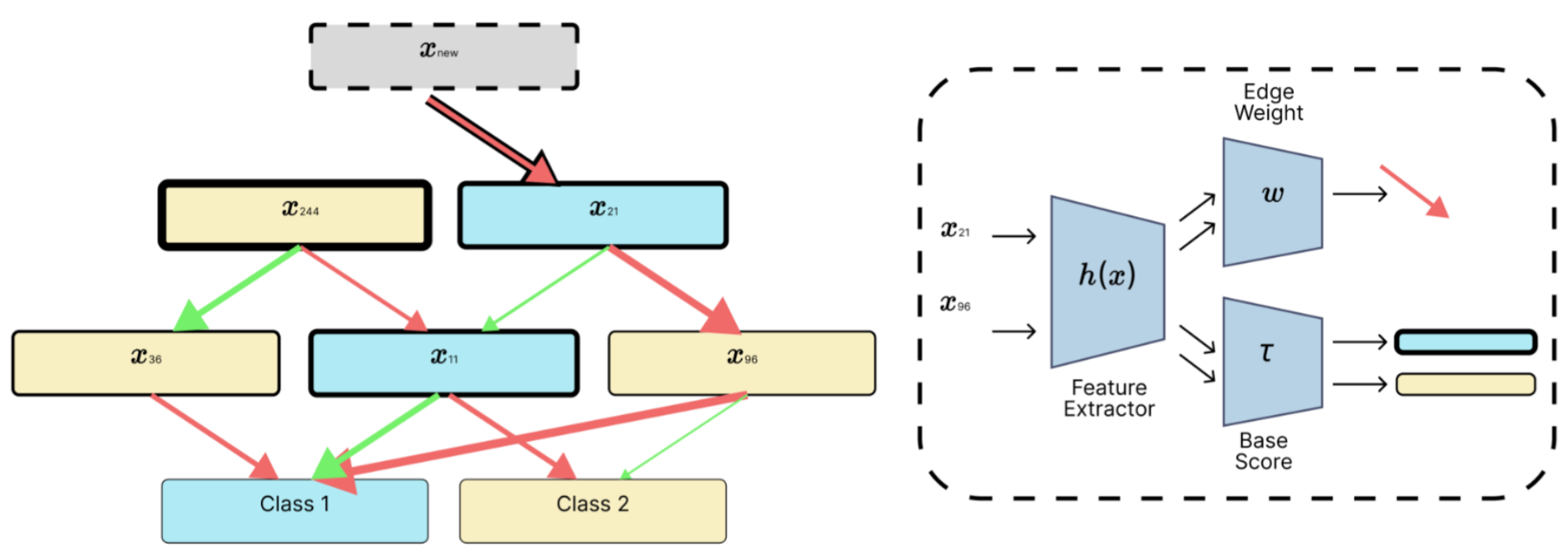
Supports in Practice (Preliminary work)

Slot Attention Argumentation for Case-Based Reasoning

Methods	Test Set			
	Accuracy	Precision	Recall	F1
SAA-CBR	75.13 $\pm$ 2.16	75.31 $\pm$ 2.10	75.13 $\pm$ 2.16	74.77 $\pm$ 2.28
w/o feature combination	54.68 $\pm$ 3.69	62.97 $\pm$ 1.53	54.68 $\pm$ 3.69	47.11 $\pm$ 3.92
w/o thresholding	68.04 $\pm$ 4.61	70.70 $\pm$ 6.00	68.04 $\pm$ 4.61	66.02 $\pm$ 4.76
w/o supports	74.30 $\pm$ 2.96	74.79 $\pm$ 2.39	74.30 $\pm$ 2.96	73.73 $\pm$ 3.44
w/o supports, thresholding, feature combination	55.16 $\pm$ 3.81	62.37 $\pm$ 2.12	55.16 $\pm$ 3.81	48.88 $\pm$ 4.76

Supports in Practice (Accepted at NeSy 25)

Neuro-Argumentative Learning with Case-Based Reasoning



Conclusion

- We introduce Supported AA-CBR, a model that uses supports to removes spikes by allowing all data points to meaningfully contribute to the classification.
- Future Work:
 - Explore how under what conditions supports empirically changes the model classification performance
 - Explore how supports behaves in AA-CBR variants such as Preference-Based AA-CBR (Gould, et al. 2024) and cumulative AA-CBR (Paulino Passos, et al. 2021).

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Thank you

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